

Decision Tree Model and Business Interpretation Based on a Business Plan

FitLife Studio Analytics: Membership Renewal Risk Prediction

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1. Introduction

This report presents the development and business interpretation of a Decision Tree classification model for **FitLife Studio Analytics**. The purpose of the model is to predict whether a fitness studio member is at risk of not renewing their membership.

The assignment requires technical implementation and business interpretation. Therefore, this report explains the business problem, dataset preparation, model training, model evaluation, feature importance, model limitations, and the business meaning of the results.

This machine learning task is a **classification problem** because the target variable has two possible classes:

- **Yes:** The member is at risk of not renewing.
 - **No:** The member is not currently considered high risk.
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2. Business Problem

FitLife Studio is a fitness subscription business that provides membership plans to its members. A significant part of the company's revenue comes from recurring membership renewals. Therefore, retaining existing members is an important business priority.

The main business problem is membership churn. Some members may become less engaged over time by taking fewer classes, visiting the studio less often, using the app less frequently, experiencing payment issues, having lower satisfaction, or not responding to renewal messages. These behaviours may indicate that a member is at risk of not renewing their membership.

The key business question is: **Can FitLife Studio predict whether a member is at risk of not renewing their membership using available member information, engagement data, satisfaction indicators, payment history, and membership details?**

3. Machine Learning Decision

The Decision Tree model can help FitLife Studio identify which members may require proactive retention support. Instead of waiting until members cancel or fail to renew, FitLife can use the model to predict which members are likely to be at risk.

Based on the model's prediction, FitLife Studio can consider actions such as:

- Sending renewal reminders
- Offering personalized class recommendations
- Contacting members with low satisfaction
- Providing billing assistance for members with payment issues
- Encouraging app engagement
- Offering retention support to high-risk members
- Prioritizing staff follow-up based on predicted renewal risk

This prediction is useful because early intervention can help FitLife Studio reduce membership churn, improve member retention, and protect recurring revenue. However, the model should

support human decision-making rather than replace it, because staff may also need to consider individual member circumstances before taking action.

4. Dataset Overview

The dataset used for this project is **fitlife_membership_renewal_risk_dataset.xlsx**.

The dataset contains information about FitLife Studio members, including demographic details, membership information, engagement behaviour, satisfaction indicators, payment issues, communication history, and renewal-risk status.

The original dataset contained **416 rows and 21 columns**. After checking for and removing duplicate rows, the cleaned dataset contained **410 rows and 21 columns**. The number of columns remained the same because duplicate removal affected only repeated records, not the variables included in the dataset.

5. Target Variable and Input Features

Target Variable

The target variable in this project is **At_Risk_Not_Renewing**. This variable indicates whether a member is at risk of not renewing their membership.

- **Yes:** The member is at risk of not renewing.
- **No:** The member is not currently considered high risk.

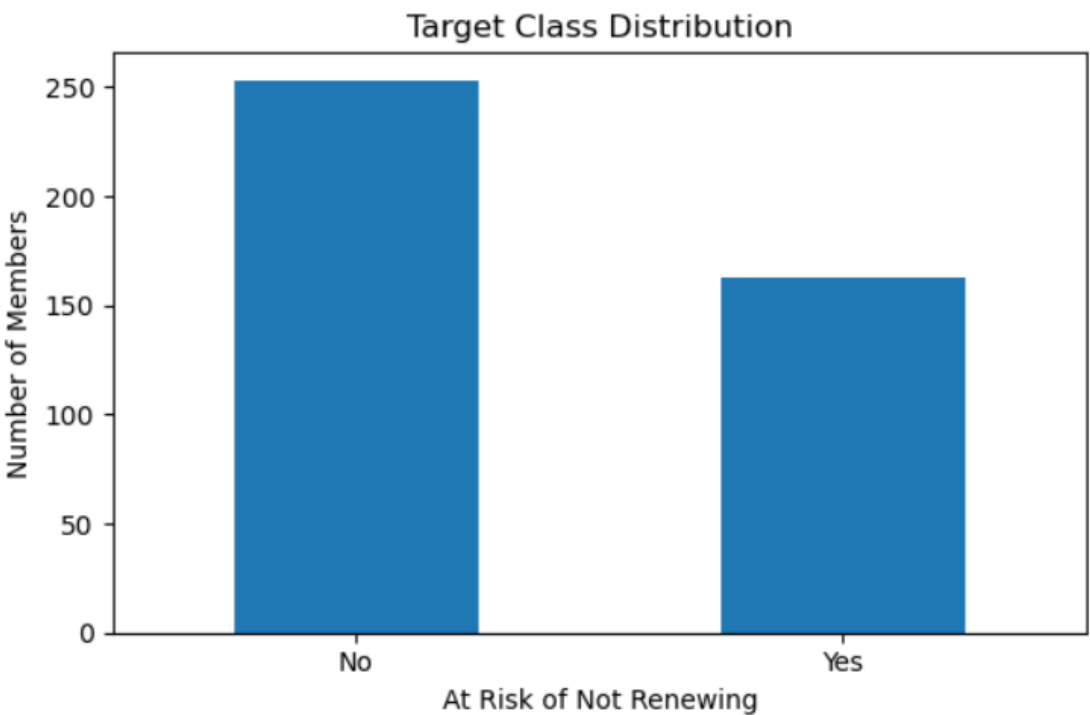


Figure 1: Target Variable Distribution

Figure 1 shows the distribution of the target variable, **At_Risk_Not_Renewing**. This helps identify how many members in the dataset are classified as at risk and not at risk of not renewing their membership.

Input Features

The input features are the variables used by the model to predict whether a member is at risk of not renewing. The dataset includes the following input features:

Age, Gender, Membership type, Monthly fee, Months as member, Contract type, Average weekly visits, Classes per month, Personal training, App login days in the last 30 days, Payment issues in the last 6 months, Satisfaction score, Complaint count in the last 3 months, Distance from studio, Last renewal discount used, Renewal reminder sent, Corporate plan, Season, Referral source

The **Member_ID** column was removed before model training because it is only an identifier. It does not represent member behaviour, engagement, satisfaction, or any meaningful business factor for predicting renewal risk.

6. Data Preparation

Data preparation was completed before model training to ensure that the dataset was clean and suitable for machine learning.

6.1 Loading and Inspecting the Dataset

The dataset was loaded into pandas. First checks were done to assess the layout and quality of the data. This included checking the number of rows and columns, viewing the first few records, checking data types, finding missing values, checking repeated rows, checking target variable values, and dividing numeric and category-based features.

This step helped confirm the structure of the dataset and identify the required data-cleaning steps before model training.

6.2 Duplicate Rows

Duplicate rows were checked and removed from the dataset. The dataset contained **6 duplicate rows** before cleaning. After removing these duplicate records, there were **0 duplicate rows** remaining.

Removing duplicates is important because repeated member records can give extra weight to the same information and may affect the model training process.

6.3 Missing Values

Missing values were identified and handled in the cleaned dataset.

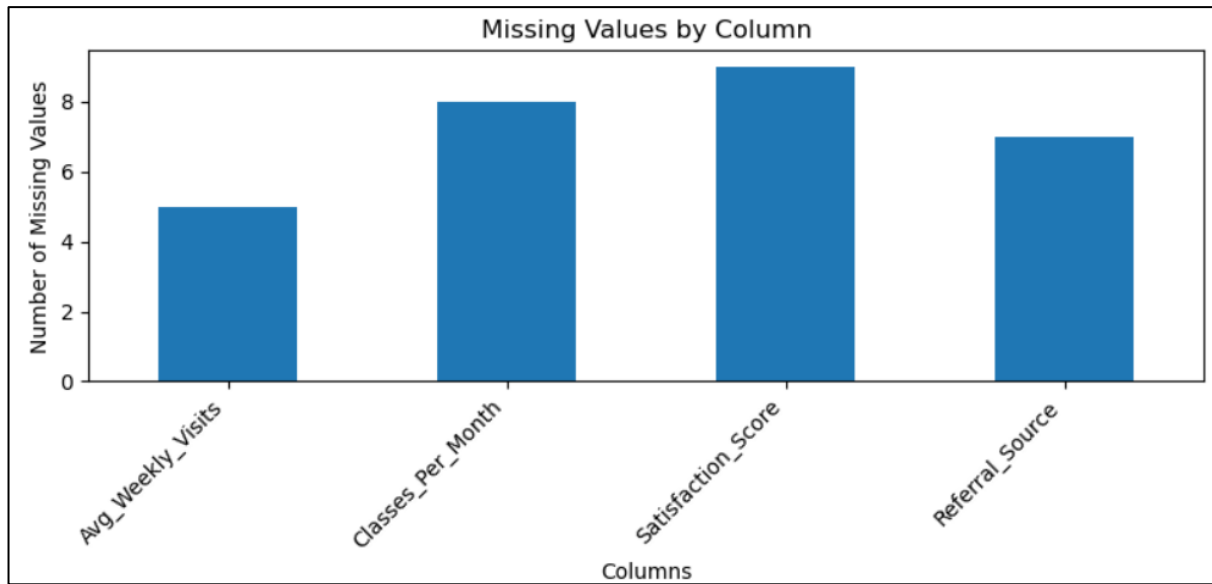


Figure 2: Missing Values by Column

Figure 2 shows the columns that contained missing values before cleaning. Missing numerical values were filled using the median, while missing categorical values were filled using the mode.

Table 1: Handling Missing Values

Column	Data Type	Missing Values Filled	Method	Fill Value
Avg_Weekly_Visits	Numerical	5	Median	2.5
Classes_Per_Month	Numerical	8	Median	3.0
Satisfaction_Score	Numerical	9	Median	7.6
Referral_Source	Categorical	7	Mode	Friend Referral

Numerical missing values were filled using the **median** because the median is less affected by extreme values. Categorical missing values were filled using the **mode**, which represents the most frequent category.

The target variable was not filled because it is the value the model is trying to predict.

7. Feature and Target Separation

After cleaning the dataset, the input features and target variable were separated.

- The feature dataset X was created by removing: At_Risk_Not_Renewing and Member_ID
- The target variable y was created using: At_Risk_Not_Renewing

This step ensured that the model learned from meaningful business variables and did not use the ID column.

8. Train-Test Split

The dataset was split into training and testing sets.

The training set was used to train the Decision Tree model, while the testing set was used to evaluate how well the model performs on unseen data.

A stratified split was used so that the class distribution of the target variable remained balanced between the training and testing sets. This is important because the model should learn from a representative sample of both at-risk and not-at-risk members.

9. Preprocessing

Machine learning models require numerical input. Since the dataset contains categorical variables, preprocessing was required.

Categorical variables were converted into numerical columns using **one-hot encoding**. Numerical variables were passed through without scaling because Decision Tree models do not require feature scaling.

The preprocessing step was fitted on the training data and applied to the testing data. This helped prepare the data for model training while keeping training and testing data separate.

10. Decision Tree Model Training

A **Decision Tree Classifier** was trained for this assignment.

The model was created using: **DecisionTreeClassifier(max_depth=4, random_state=42)**. The max_depth=4 setting was used to keep the tree reasonably simple and interpretable. This helps reduce the risk of overfitting and makes the model easier to explain from a business perspective.

The model was trained using the processed training data.

11. Model Predictions

After training, the model was used to make predictions on the testing data. The predicted values were compared with the actual target values to evaluate model performance.

The model predicted whether each member in the testing set was:

- at risk of not renewing, or
- not currently at high risk.

This comparison allowed the model to be evaluated using accuracy, precision, recall, F1-score, and the confusion matrix.

12. Model Evaluation Results

The Decision Tree model was evaluated using the testing data.

Table 2: Decision Tree Evaluation Metrics

Metric	Score
Accuracy	0.731707
Precision for Yes	0.692308
Recall for Yes	0.562500
F1-Score for Yes	0.620690

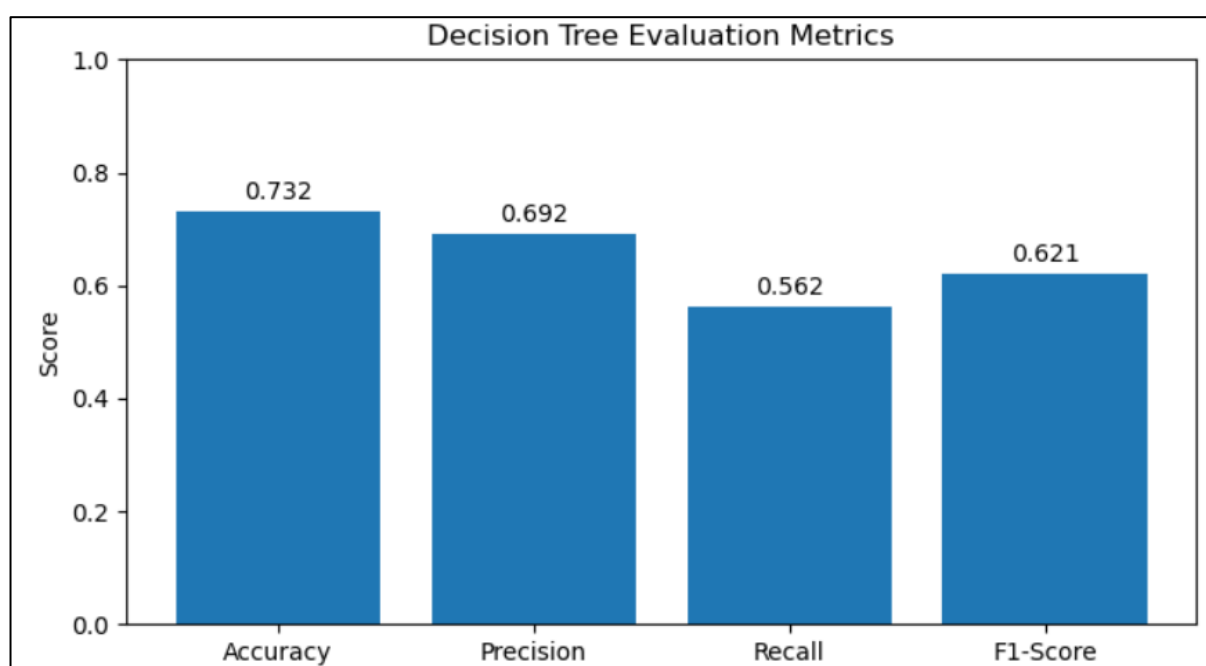


Figure 3: Decision Tree Evaluation Metrics

This figure compares the Decision Tree model's accuracy, precision, recall, and F1-score. The chart shows that the model has reasonable overall accuracy, while recall for the at-risk class is moderate.

12.2 Interpretation of Metrics

Accuracy measures the overall percentage of correct predictions. The model achieved a testing accuracy of approximately **73.17%**, meaning it correctly classified about 73% of members in the testing dataset.

Precision for the Yes class measures how many members predicted as at risk were actually at risk. The precision score was approximately **69.23%**, meaning that when the model predicted a member as at risk, it was correct about 69% of the time.

Recall for the Yes class measures how many actual at-risk members the model successfully identified. The recall score was approximately **56.25%**, meaning the model identified just over half of the actual at-risk members.

F1-score balances precision and recall. The F1-score was approximately **62.07%**, which indicates moderate performance for the at-risk class.

For this business problem, recall for the **Yes** class is especially important because missing at-risk members means FitLife may lose the opportunity to take proactive retention action.

13. Training Accuracy, Testing Accuracy, and Overfitting

The model's training accuracy and testing accuracy were compared to check whether the Decision Tree may be overfitting.

Table 3: Training and Testing Accuracy

Evaluation Set	Accuracy
Training Set	0.780488
Testing Set	0.731707

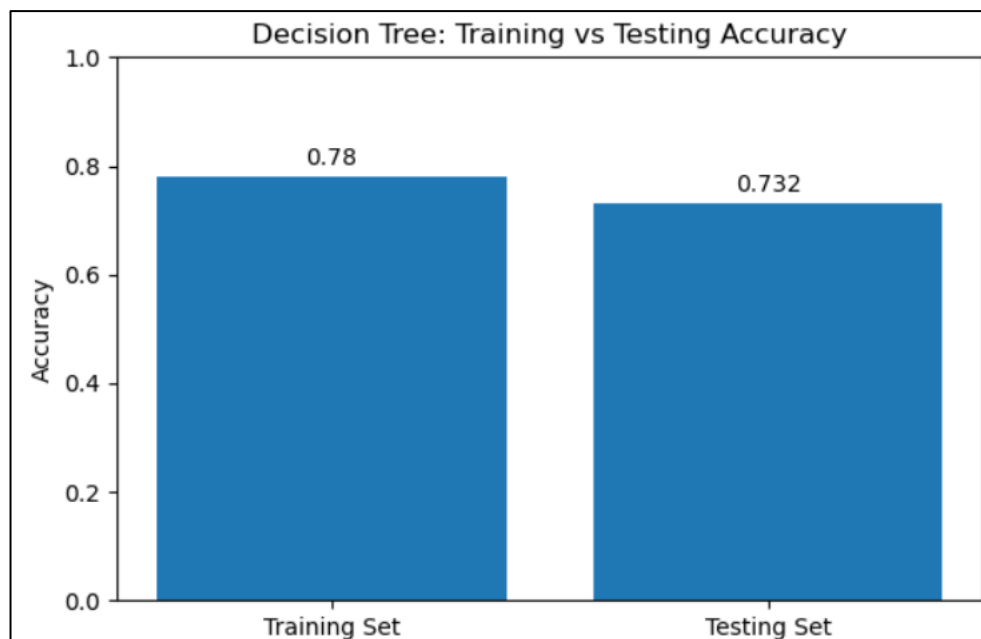


Figure 4: Training vs Testing Accuracy Chart

This figure compares the training and testing accuracy of the Decision Tree model. The small difference between the two values suggests that the model does not show strong signs of overfitting.

The difference between training accuracy and testing accuracy was approximately **0.048780**.

This difference is relatively small. Therefore, the model does not show strong signs of overfitting. The model appears to learn useful patterns from the training data while still performing reasonably well on unseen testing data.

If training accuracy were much higher than testing accuracy, that would suggest that the model memorized the training data rather than learning general patterns. In this case, the accuracy gap is not large, so the model is reasonably balanced.

14. Confusion Matrix and Business Meaning

The confusion matrix provides a more detailed view of correct and incorrect predictions.

Table 4: Decision Tree Confusion Matrix for Membership Renewal Risk Prediction

	Predicted No	Predicted Yes
Actual No	42	8
Actual Yes	14	18

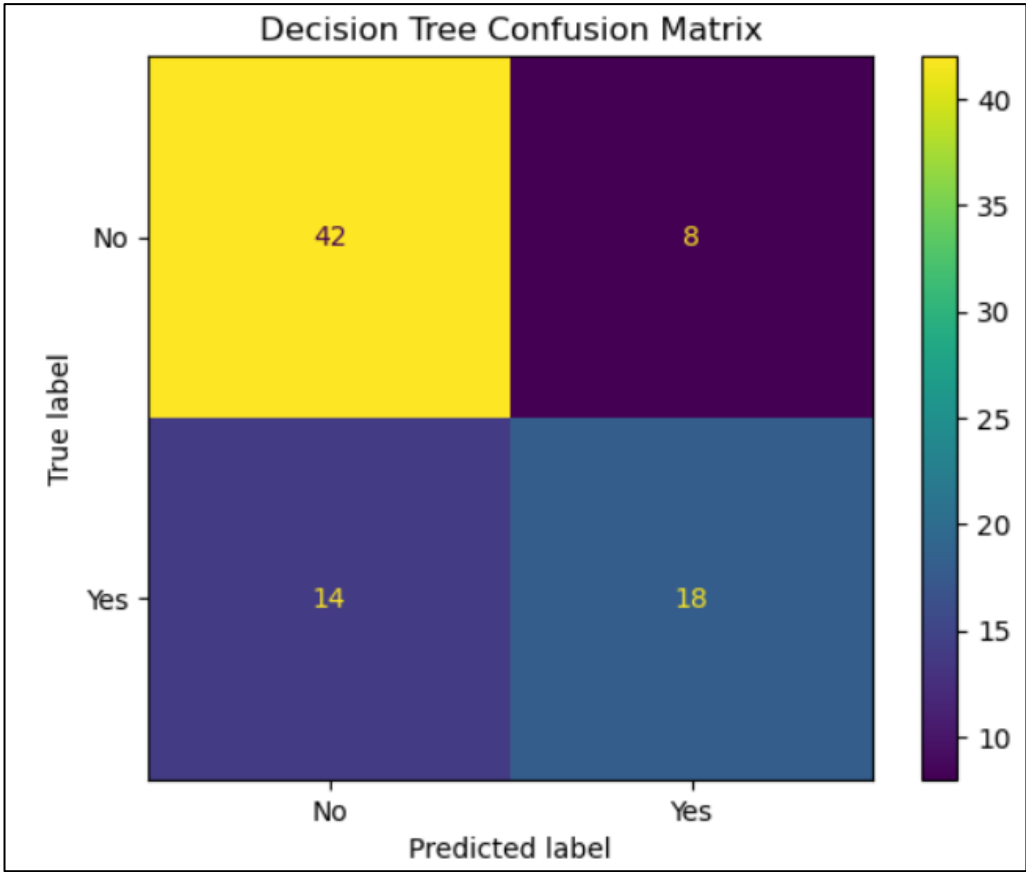


Figure 5: Confusion Matrix Chart

The confusion matrix shows the model’s correct and incorrect predictions. In the business context, false negatives are especially important because they represent at-risk members who were missed by the model.

Interpretation

- The model correctly predicted **42 members** as not at risk. These are true negatives.
- The model correctly predicted **18 members** as at risk. These are true positives.
- The model incorrectly predicted **8 members** as at risk when they were not actually at risk. These are false positives.
- The model incorrectly predicted **14 members** as not at risk when they were actually at risk. These are false negatives.

False Positives in Business Context

A false positive means the model predicts that a member is at risk of not renewing, but the member is actually not at risk.

In business terms, FitLife may spend extra time or resources contacting members who may not need retention support. This may create some inefficiency, but it is usually less harmful than missing members who are truly at risk.

False Negatives in Business Context

A false negative means the model predicts that a member is not at risk, but the member is actually at risk of not renewing.

This is more serious for FitLife because the business may fail to contact a member who needs retention support. As a result, FitLife may lose an opportunity to prevent membership churn.

For this reason, recall for the at-risk class is an important metric.

15. Feature Importance

Decision Trees can identify which features were most important when making predictions. In this model, feature importance helps explain which variables the Decision Tree relied on most when classifying whether a member is at risk of not renewing their membership.

The top 10 most important features identified by the model were:

1. Average weekly visits
2. Classes per month
3. Monthly contract type
4. Satisfaction score
5. Renewal reminder sent
6. Basic membership type
7. App login days in the last 30 days
8. Payment issues in the last 6 months
9. Premium membership type
10. Age

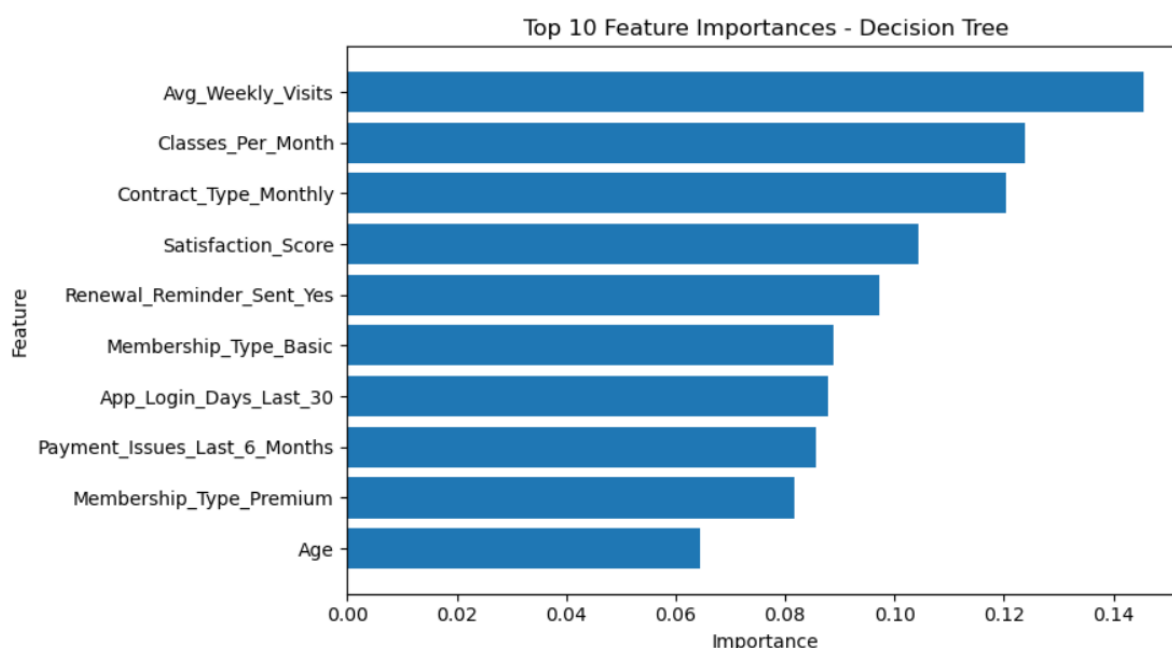


Figure 6: Top 10 Feature Importances Chart

Figure 6 shows the top 10 features used by the Decision Tree model. The results suggest that member engagement, satisfaction, contract type, renewal reminders, payment issues, membership type, and age were useful indicators for predicting renewal risk.

The most important features, such as average weekly visits, classes per month, and app login days, are related to member engagement. This means members who visit less often, attend fewer classes, or use the app less may have a higher risk of not renewing their membership.

Some features, such as monthly contract type and membership type categories, were created through one-hot encoding. This means the original categorical variables were converted into separate binary features before training the model. The Decision Tree found some of these specific categories useful for predicting renewal risk.

Overall, the feature importance results are useful for business decision-making because they show which factors FitLife Studio should monitor closely. For example, members with low engagement, low satisfaction, payment issues, or monthly contracts may need targeted retention actions such as follow-up messages, special offers, or personalized support.

16. Business Interpretation of Important Features

The feature importance results provide practical insights for FitLife Studio. The most important features show that renewal risk is strongly connected to member engagement, membership structure, satisfaction, communication, and payment experience.

Average Weekly Visits and Classes Per Month

Average weekly visits and classes per month were among the most important features in the Decision Tree model. These variables reflect member engagement. Members who visit the studio

less often or attend fewer classes may not be receiving enough value from their membership, which can increase the risk of not renewing.

FitLife can use these indicators to identify members with declining engagement. For example, members with low visit frequency or reduced class participation can be contacted with personalized class recommendations, fitness goal check-ins, or encouragement to participate in group programs.

Contract Type and Membership Type

Contract type and membership type were also important predictors of renewal risk. This suggests that members with different membership arrangements may show different renewal behaviours. For example, monthly members may have more flexibility to stop renewing compared with members on longer-term contracts.

FitLife can use this insight to design different retention strategies for monthly, six-month, annual, basic, standard, and premium members. Members with more flexible plans may need earlier renewal reminders or stronger engagement support.

Satisfaction Score

Satisfaction score is directly related to the member experience. A lower satisfaction score may indicate that a member is unhappy with the service, facilities, class options, or overall membership value. This can increase the likelihood of non-renewal.

FitLife can use satisfaction scores to identify members who may need service recovery. Staff can follow up with dissatisfied members, understand their concerns, and offer personalized support before the renewal period.

Renewal Reminder Sent

Renewal reminder status was also useful in the model. This suggests that communication may influence renewal behaviour. Members who do not receive or respond to renewal reminders may be more likely to miss renewal opportunities or become inactive.

FitLife can improve its reminder strategy by sending timely renewal messages through email, app notifications, phone calls, or in-person follow-ups. Reminder messages can also be personalized based on each member's engagement level and membership type.

Payment Issues

Payment issues in the last six months were another important feature. Payment problems may create inconvenience for members and may also indicate billing difficulties or financial concerns.

FitLife can use this information to provide billing support before the renewal period. By helping members resolve payment issues early, FitLife may reduce friction and improve the chance of renewal.

Overall, these important features can help FitLife move from a reactive retention approach to a proactive retention strategy. Instead of waiting for members to stop renewing, FitLife can monitor engagement, satisfaction, contract type, reminders, and payment issues to identify members who may need support earlier.

The following sections directly interpret the model results in relation to the business questions required in the assignment.

17. Overall Business Interpretation

The Decision Tree model performed reasonably well for a first business classification model. A testing accuracy of approximately **73.17%** suggests that the model can correctly classify a meaningful portion of members.

However, the model does not identify all at-risk members. The recall score of approximately **56.25%** means that some at-risk members are missed. Since FitLife's goal is to reduce membership churn, the model should be used as a screening tool rather than a final decision-making system. For this business problem, recall for the Yes class is the most important metric because FitLife wants to identify as many truly at-risk members as possible before they fail to renew.

A false negative is more costly than a false positive because a missed at-risk member may not receive timely retention support.

The model is useful because it highlights which members may need attention and which features are most related to renewal risk. FitLife can use this information to improve retention planning and customer relationship management.

18. Limitation and Possible Bias

One limitation of the model is that it is based only on the available dataset. The dataset may not include all reasons why a member chooses not to renew. For example, a member may leave because of relocation, personal schedule changes, family reasons, health concerns, competitor offers, or financial constraints. These factors may not be captured in the dataset.

Another limitation is that the dataset is relatively small. With more member records, the model may learn stronger and more reliable patterns.

A possible bias may also exist if certain groups of members are underrepresented in the dataset. If the dataset does not reflect all types of FitLife members fairly, the model's predictions may be less accurate for some groups.

19. Why Human Judgment Is Still Needed

Human judgment should still be used when making business decisions based on the model results.

The model can identify patterns in the dataset, but it cannot fully understand personal circumstances. For example, a member may reduce visits temporarily because of work, travel, illness, family responsibilities, or schedule conflicts. The model may interpret this as risk, even if the member still intends to renew.

FitLife managers and staff should use the model as a decision-support tool. They should combine the model's prediction with customer service knowledge, member communication history, and professional judgment.

Human review is also important to ensure that members are treated fairly and respectfully. The model should support better decisions, not replace human decision-making.

20. Final Conclusion

This project developed a Decision Tree classification model to predict whether FitLife Studio members are at risk of not renewing their membership.

The model achieved approximately **73.17% testing accuracy**, with a precision of **69.23%**, recall of **56.25%**, and F1-score of **62.07%** for the at-risk class.

The training and testing accuracy comparison showed that the model does not have strong signs of overfitting. The model is reasonably balanced and interpretable.

The most important features included average weekly visits, classes per month, contract type, satisfaction score, renewal reminders, membership type, app login activity, payment issues, and age.

From a business perspective, the model can help FitLife identify members who may require proactive retention support. However, because the model misses some actual at-risk members, it should be used together with human judgment and business knowledge.

Overall, the Decision Tree model provides useful support for membership renewal risk prediction and can help FitLife improve its retention strategy.